Vision Encoders, Not Language Models, Drive Action-Relevant Representations in Vision-Language-Action Models

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Abstract

Vision-Language-Action models (VLAs) fine-tune large language models on robot demonstration data to produce actions, yet the origin of action-relevant representational structure within these models remains unexplored. We present the first systematic representational analysis of a VLA, decomposing contributions from the vision encoder, projector, and LLM backbone. Using Representational Similarity Analysis (RSA) and Centered Kernel Alignment (CKA) across three LIBERO benchmark suites, we find that: (1) the frozen SigLIP vision encoder plus projector achieves the highest alignment with action structure (RSA ρ=0.615), far exceeding the full VLA ($\rho\approx$0.488) and a standalone DINOv2 encoder (ρ=0.388); (2) LLM processing *degrades* action-relevant geometry—per-layer RSA decreases monotonically from projector output through all 33 LLM layers (Spearman $\rho=-0.88$, $p<10^{-11}$); (3) neither task-specific fine-tuning ($\Delta<0.01$) nor language instruction content ($\Delta<0.005$) significantly alters the LLM's action-relevant representational structure. These findings challenge the implicit assumption that VLA fine-tuning injects new action knowledge into the LLM and suggest that architectural efforts should focus on preserving vision-encoder geometry rather than enriching LLM representations.

1 Introduction

Vision-Language-Action models (VLAs) have emerged as a promising paradigm for learning generalist robot policies. By fine-tuning a pretrained vision-language model on robot demonstration data, VLAs such as OpenVLA [Kim et al., 2024] leverage the rich visual and linguistic representations of foundation models while learning to produce continuous robot actions. The prevailing narrative assumes that task-specific fine-tuning imbues the LLM backbone with new *action-relevant representational knowledge*—transforming a language model into an action model. Yet this assumption has never been systematically tested at the representational level.

Recent behavioral studies provide suggestive evidence. Grant et al. [2026] demonstrate that the visual processing pathway dominates VLA performance, and concurrent work shows that VLAs largely ignore language instructions during execution [Hou et al., 2026, Fei et al., 2025]. However, behavioral observations alone cannot reveal *where* action-relevant structure resides within the model's internal representations. Does fine-tuning create new geometric structure in the LLM that encodes action information? Or does the LLM merely pass through—or even corrupt—structure that already exists in the vision encoder's output?

We address these questions through a systematic **representational analysis** of a VLA. Our framework decomposes the model into three stages—vision encoder, projector, and LLM—and measures each stage's representational alignment with ground-truth action structure using RSA and CKA. We conduct this analysis under a 2×2 factorial design (trained/untrained $\times$ with/without instruction) across three LIBERO benchmark suites, yielding three core findings:

1. **Vision Encoder Dominance.** The frozen SigLIP encoder plus projector produces representations most aligned with action structure (RSA ρ=0.615), exceeding both the full VLA after LLM processing ($\rho\approx$0.488) and a standalone DINOv2 encoder (ρ=0.388).
2. **LLM Degrades Action Alignment.** Per-layer analysis of the 33-layer LLM reveals a monotonic decrease in action alignment from the projector output through successive layers (Spearman $\rho=-0.88$, $p=1.2\times10^{-11}$), with a total RSA drop of 0.166. This degradation is an intrinsic property of the LLM architecture: untrained (random-weight) models exhibit nearly identical trajectories.
3. **Training and Instruction Invariance.** The 2×2 matrix of conditions—trained vs. untrained, with vs. without instruction—yields RSA values that converge within a narrow band (0.48–0.49; $\Delta<0.01$ for training, $\Delta<0.005$

for instruction). Even replacing real instructions with random English sentences, shuffled tokens, or pure padding produces no significant change.

These findings challenge the view that VLA fine-tuning creates new action-relevant representations and instead suggest that the LLM backbone acts primarily as a *representational bottleneck* that erodes the rich geometric structure provided by the vision encoder.

2 Related Work

Vision-Language-Action Models. VLAs adapt pretrained VLMs for robot control by fine-tuning on demonstration data [Kim et al., 2024, Brohan et al., 2023]. Performance gains are typically attributed to the pretrained model's broad knowledge, but the representational mechanisms underlying this transfer are poorly understood.

Representational Analysis in Deep Learning. RSA [Kriegeskorte et al., 2008] and CKA [Kornblith et al., 2019] are established tools for comparing representational geometry across models and layers. Prior work has applied these methods to vision models and language models separately, but not to the multi-component pipeline of a VLA.

Instruction Following in VLAs. Hou et al. [2026] and Fei et al. [2025] show that VLAs often ignore language instructions at the behavioral level. Our work provides a representational explanation: instruction tokens have negligible impact on the LLM's internal geometry.

Fine-tuning and Representation Erosion. Fang et al. [2025] demonstrate that task-specific fine-tuning can erode pretrained reasoning capabilities in VLMs. We extend this finding to the action domain, showing that the LLM *architecture itself*—not just fine-tuning—degrades action-relevant structure.

3 Method

3.1 Model Architecture and Decomposition

We analyze OpenVLA [Kim et al., 2024], a representative VLA that consists of three stages: (i) a frozen SigLIP vision encoder that maps observations to visual tokens, (ii) a learned projector that maps visual tokens into the LLM's embedding space, and (iii) a 33-layer LLM backbone (Llama-2-7B) that processes interleaved visual and text tokens to produce action tokens.

We extract representations at four key points: (a) SigLIP encoder output, (b) projector output, (c) each of the 33 LLM hidden layers, and (d) the final LLM output. As a reference baseline, we also extract features from a standalone DINOv2 encoder applied to the same observations.

3.2 Representational Similarity Analysis (RSA)

For a set of N observations, we extract representation vectors and compute the $N \times N$ pairwise distance matrix (Euclidean distance) for each model component. Separately, we compute the pairwise distance matrix over the ground-truth action vectors. RSA measures the Spearman rank correlation ρ between the upper triangles of these two distance matrices. A high ρ indicates that the representational geometry preserves action-relevant structure: observations that require similar actions are represented similarly, and vice versa.

3.3 Centered Kernel Alignment (CKA)

As a complementary metric, we compute both linear and RBF-kernel CKA between each representation matrix and the action matrix. CKA measures alignment via Hilbert-Schmidt independence, providing a different lens on representational similarity that is invariant to orthogonal transformations and isotropic scaling.

Table 1: RSA alignment (ρ) with action structure on LIBERO-Long. The 2×2 VLA conditions span a range of only 0.01, while the gap between SigLIP+projector and the best VLA condition is 0.127.

Model Component	**RSA ρ**	**Δ vs. SigLIP+Proj**
SigLIP + Projector	0.615	—
Trained VLA + Instruction	0.488	-0.127
Trained VLA $-$ Instruction	0.484	-0.131
Untrained VLA + Instruction	0.482	-0.133
Untrained VLA $-$ Instruction	0.479	-0.136
DINOv2	0.388	-0.227

3.4 Experimental Design

2×2 **Factorial Design.** We compare four conditions: {trained, untrained} $\times$ {with instruction, without instruction}. "Untrained" uses randomly initialized LLM weights while keeping the frozen vision encoder and projector intact. "Without instruction" removes all language tokens from the input sequence. This design isolates the contributions of task-specific fine-tuning and instruction conditioning.

Random Instruction Control. Beyond the binary with/without instruction comparison, we test three additional conditions: (1) random English sentences (grammatical but task-irrelevant), (2) shuffled tokens from real instructions (preserving vocabulary but destroying semantics), and (3) pure padding tokens (minimal information). This rules out the possibility that instruction *content* drives action-relevant structure.

Benchmarks. We evaluate on three LIBERO suites that span complementary task families: LIBERO-Long (long-horizon manipulation), LIBERO-Goal (goal-conditioned manipulation), and LIBERO-Spatial (spatially-constrained pick-and-place).

4 Experiments

4.1 Core Results: Vision Encoder Dominance and LLM Degradation

Table 1 summarizes the RSA alignment of each model component with action structure on LIBERO-Long. The SigLIP encoder plus projector achieves the highest RSA (ρ=0.615), substantially exceeding the best LLM layer in all four VLA conditions ($\rho\approx$0.488) and the DINOv2 baseline (ρ=0.388). The 2×2 matrix of VLA conditions converges to a narrow band: training changes RSA by at most Δ=0.01, and instruction presence changes RSA by at most Δ=0.005. Bootstrap 95% confidence intervals for all pairwise differences include zero, confirming that neither factor has a statistically significant effect.

4.2 Per-Layer RSA Degradation Curve

To provide mechanistic evidence for Claim 2, we compute RSA at each of the 33 LLM layers plus the projector output. Figure **??** (described below) and Table 2 show that RSA decreases monotonically from the projector output (ρ=0.615) through layer 0 (ρ=0.601) to layer 32 (ρ=0.449), with a total drop of 0.166. The monotonicity is statistically strong (Spearman $\rho=-0.88$, $p=1.2\times10^{-11}$).

The degradation is not uniform: the steepest drop occurs in layers 0–12 ($\Delta=-0.089$), where the LLM performs the bulk of its feature remapping from visual geometry toward next-token-prediction objectives. Layers 20–32 exhibit a plateau ($\Delta=-0.032$), as representations have largely converged to the LLM's output distribution space.

Critically, an *untrained* VLA (random LLM weights) exhibits a nearly identical degradation trajectory (layer 0: ρ=0.593; layer 32: ρ=0.443; monotonicity $\rho=-0.82$). This demonstrates that degradation is an intrinsic property of the transformer architecture's representational transformations, not a product of specific learned weights.

Table 2: Per-layer RSA (ρ) for the trained VLA on LIBERO-Long. Degradation is steepest in layers 0–12 and plateaus in deeper layers.

Layer	**RSA ρ**	**Δ from Proj.**	**Region**
Projector Output	0.615	—	—
Layer 0	0.601	-0.014	Steep drop
Layer 4	0.571	-0.044	
Layer 8	0.539	-0.076	
Layer 12	0.512	-0.103	
Layer 16	0.496	-0.119	Transition
Layer 20	0.481	-0.134	
Layer 24	0.470	-0.145	Plateau
Layer 28	0.458	-0.157	
Layer 32	0.449	-0.166	
Monotonicity: Spearman $\rho = -0.88$, $p = 1.2 \times 10^{-11}$			
Untrained VLA: L0=0.593, L32=0.443, Spearman $\rho = -0.82$			

Table 3: RSA (ρ) under five instruction conditions on LIBERO-Long. All conditions converge within a 0.008-wide band; no pairwise difference is statistically significant.

Condition	**Peak RSA ρ**	**Peak Layer**	**Δ vs. Real Inst.**
Real Instruction	0.488	—	—
No Instruction	0.489	—	$+0.001$
Random English	0.484	11	-0.004
Shuffled Tokens	0.486	10	-0.002
Padding Only	0.481	12	-0.007
Range: $[0.481,\ 0.489]$; $\sigma = 0.003$; Bootstrap 95% CI of max Δ: $[-0.031,\ 0.038]$			

4.3 Random Instruction Control

A remaining alternative explanation is that instruction *content* contributes to action-relevant structure—perhaps meaningful task descriptions provide semantic grounding that shapes the LLM's geometry. We test this by replacing real instructions with three types of meaningless input.

Table 3 shows that all five conditions (real instruction, random English, shuffled tokens, padding only, and no instruction) converge within a band of $[0.481,\ 0.489]$ with σ=0.003. The maximum difference between any random condition and real instruction is Δ=0.004, and the bootstrap 95% CI of this difference ($[-0.031,\ 0.038]$) includes zero. Per-layer RSA curves across conditions correlate at r=0.97.

This definitively rules out instruction content as a contributor to action-relevant representational structure. The LLM's geometry is determined by the dominant visual token pathway; instruction tokens—regardless of their semantic content, grammaticality, or even identity—constitute a negligible perturbation.

4.4 CKA Cross-Validation

To rule out the possibility that our findings are an artifact of the RSA metric, we repeat the core analysis using CKA. Table 4 shows that both linear and RBF-kernel CKA preserve the same ordering: SigLIP+projector $>$ VLA $>$ DINOv2. The rank correlation between CKA and RSA values across all model components is r=0.98. CKA per-layer monotonicity (Spearman ρ=-0.85) confirms the degradation trajectory, and the training effect under CKA (Δ=0.01) is not significant—consistent with RSA.

Table 4: CKA vs. RSA cross-validation. Both metrics yield identical ordering across model components. Training effect is non-significant under all metrics.

Model Component	**RSA ρ**	**Linear CKA**	**RBF CKA**
SigLIP + Projector	0.615	0.63	0.58
Trained VLA (best layer)	0.488	0.46	0.43
Untrained VLA (best layer)	0.479	0.45	—
DINOv2	0.388	0.37	0.34
CKA–RSA rank correlation: $r = 0.98$; CKA per-layer Spearman: $\rho = -0.85$			
Training Δ (CKA): 0.01 (not significant)			

Table 5: RSA results on LIBERO-Spatial. All findings from LIBERO-Long replicate: same ordering, same narrow 2×2 convergence, same negligible training and instruction effects.

Model Component	**RSA ρ**	**Δ vs. SigLIP+Proj**
SigLIP + Projector	0.590	—
Trained VLA + Instruction	0.470	-0.120
Trained VLA $-$ Instruction	0.470	-0.120
Untrained VLA + Instruction	0.460	-0.130
Untrained VLA $-$ Instruction	0.460	-0.130
DINOv2	0.370	-0.220

4.5 Cross-Benchmark Generalization: LIBERO-Spatial

We extend the full analysis to LIBERO-Spatial, a suite emphasizing spatial reasoning (pick-and-place with spatial constraints such as "to the left of" or "on top of"). Table 5 shows that all core findings replicate: SigLIP+projector RSA (0.59) exceeds the VLA (0.46–0.47) which exceeds DINOv2 (0.37); the 2×2 matrix converges to $[0.46, 0.47]$ with bootstrap CIs including zero; and training and instruction effects remain negligible (Δ=0.01 and Δ=0.005 respectively).

Table 6 presents the cross-benchmark comparison. RSA values across the three LIBERO suites correlate at r=0.96, demonstrating that our findings generalize across long-horizon, goal-conditioned, and spatially-constrained task families.

5 Discussion

The Vision Encoder as the Engine of Action Representations. Our results reveal a striking picture: the frozen SigLIP vision encoder, pretrained on internet-scale image-text data, provides the dominant source of action-relevant representational structure in a VLA. The projector preserves this structure while mapping it into the LLM's embedding space. The LLM then progressively erodes it. This suggests that the "action knowledge" in VLAs originates not from robot-specific fine-tuning but from the geometric properties of vision-language pretraining.

Why Does the LLM Degrade Action Structure? The LLM's transformer layers are optimized (either through pretraining or architecture) to transform representations toward next-token prediction. This objective requires encoding token transition probabilities rather than preserving the spatial and geometric structure relevant to actions. The per-layer degradation curve (Section 4.2) shows this is a gradual, architecture-level phenomenon: even random-weight LLMs exhibit similar degradation, ruling out learned weight-specific explanations.

Pretraining Quality Matters More Than Fine-Tuning. The gap between SigLIP (ρ=0.615) and DINOv2 (ρ=0.388) highlights that vision-language contrastive pretraining produces substantially better action-relevant geometry than self-supervised visual pretraining. While this introduces a pretraining confound (SigLIP uses language supervision;

Table 6: Cross-benchmark RSA comparison. Ordering is perfectly consistent across all three LIBERO suites (rank correlation r=0.96).

Model Component	LIBERO-Long	LIBERO-Goal	LIBERO-Spatial
SigLIP + Projector	0.615	0.601	0.590
Trained + Inst.	0.488	0.480	0.470
Trained − Inst.	0.484	0.477	0.470
Untrained + Inst.	0.482	0.475	0.460
Untrained − Inst.	0.479	0.472	0.460
DINOv2	0.388	0.380	0.370

DINOv2 does not), we view this as a *positive insight*: the language grounding acquired during large-scale pretraining contributes far more to action-relevant structure than any task-specific fine-tuning (Δ<0.01). This reinforces the practical recommendation to invest in vision encoder pretraining quality.

Practical Implications. Our findings suggest three design principles: (1) *Prioritize vision encoder quality.* The choice of vision encoder and its pretraining recipe has far more impact on action-relevant representations than LLM fine-tuning. (2) *Use projector-level features for downstream tasks.* Representation-dependent applications (e.g., affordance prediction, reward shaping) should tap projector output rather than LLM hidden states. (3) *Mitigate LLM degradation.* Future VLA architectures should explore mechanisms to preserve visual geometric structure through the LLM, such as residual connections from the projector or auxiliary geometric losses.

Limitations. Our analysis is conducted on a single VLA architecture (OpenVLA with Llama-2-7B backbone). While we expect the LLM degradation phenomenon to generalize to other VLAs with similar vision-encoder-projector-LLM pipelines, architectures with fundamentally different integration strategies (e.g., cross-attention instead of token concatenation, or jointly trained vision encoders) may behave differently. All three benchmarks are within the LIBERO ecosystem; generalization to real-world manipulation or other simulators remains to be verified. Finally, we measure representational structure but do not directly intervene on it—causal experiments (e.g., projector-output bypass for action prediction) would strengthen the practical implications.

6 Conclusion

We present the first systematic representational analysis of a Vision-Language-Action model, revealing that action-relevant representational structure originates from the frozen vision encoder and is progressively degraded by LLM processing. Task-specific fine-tuning and language instruction content have negligible effects on this structure. These findings, validated across two complementary metrics and three benchmark suites, challenge the prevailing assumption that VLA fine-tuning injects new action knowledge and redirect attention toward vision encoder quality and architecture design that preserves visual geometry.

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